

Inverse of a Matrix

An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C such that

$$\begin{array}{ccc} CA = I & \text{and} & AC = I \\ \uparrow & & \uparrow \\ A^{-1} & & A^{-1} \end{array}$$

THEOREM 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

Elementary Matrices

An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix. The next example illustrates the three kinds of elementary matrices.

EXAMPLE 5

Let $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$,
 $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

Compute E_1A , E_2A , and E_3A , and describe how these products can be obtained by elementary row operations on A .

SOLUTION Verify that

$$E_1A = \begin{bmatrix} a & b & c \\ d & e & f \\ g - 4d & h - 4e & i - 4f \end{bmatrix}, \quad E_2A = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix},$$

$$E_3A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}.$$

Find A^{-1}

$$[A | I] \xrightarrow[E_1 E_2 \dots E_n]{RR} [I | A^{-1}]$$

Characterizations of Invertible Matrices

THEOREM 8

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

- A is an invertible matrix.
- A is row equivalent to the $n \times n$ identity matrix.
- A has n pivot positions.
- The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

Partitioned Matrices

EXAMPLE 1 The matrix

$$A = \left[\begin{array}{ccc|cc|c} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ -8 & -6 & 3 & 1 & 7 & -4 \end{array} \right]$$

can also be written as the 2×3 **partitioned** (or **block**) matrix

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

whose entries are the *blocks* (or *submatrices*)

$$A_{11} = \begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix}, \quad A_{13} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} -8 & -6 & 3 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 1 & 7 \end{bmatrix}, \quad A_{23} = \begin{bmatrix} -4 \end{bmatrix}$$

• Addition & scalar multiplication are as ordinary sum and scalar multiplication

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}+B_{11} & A_{12}+B_{12} \\ A_{21}+B_{21} & A_{22}+B_{22} \end{bmatrix}$$

$$c \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} cA_{11} & cA_{12} \\ cA_{21} & cA_{22} \end{bmatrix}$$

Multiplication of Partitioned Matrices

Partitioned matrices can be multiplied by the usual row–column rule as if the block entries were scalars, provided that for a product AB , the column partition of A matches the row partition of B .

EXAMPLE 3 Let

$$A = \left[\begin{array}{ccc|cc} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{array} \right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{array} \right] = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

The 5 columns of A are partitioned into a set of 3 columns and then a set of 2 columns. The 5 rows of B are partitioned in the same way—into a set of 3 rows and then a set of 2 rows. We say that the partitions of A and B are **conformable** for **block multiplication**. It can be shown that the ordinary product AB can be written as

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \\ 2 & 1 \end{bmatrix}$$

It is important for each smaller product in the expression for AB to be written with the submatrix from A on the left, since matrix multiplication is not commutative. For instance,

$$\begin{aligned} A_{11}B_1 &= \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} \\ A_{12}B_2 &= \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} \end{aligned}$$

Hence the top block in AB is

$$A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \end{bmatrix}$$



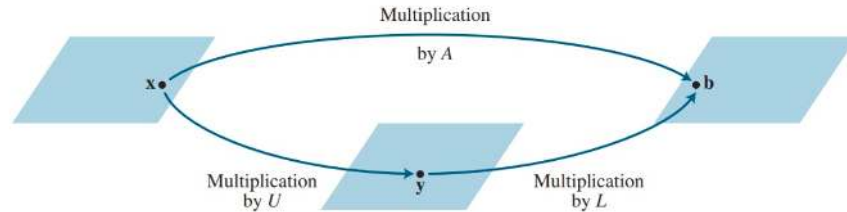
Matrix Factorization

LU Factorization:

Let matrix A be an $m \times n$ matrix, the LU factorization gives a way to rewrite $A = LU$ where L is an $m \times m$ lower triangular matrix and U is an $m \times n$ echelon form of A

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix} \begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

L U



Using LU factorization to solve: ① $L\vec{y} = \vec{b}$: $[L \vec{b}] \longrightarrow [I \vec{y}]$
② $U\vec{x} = \vec{y}$

Subspaces of $\mathbb{R}^n$

A **subspace** of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

- The zero vector is in H .
- For each $\mathbf{u}$ and $\mathbf{v}$ in H , the sum $\mathbf{u} + \mathbf{v}$ is in H .
- For each $\mathbf{u}$ in H and each scalar c , the vector $c\mathbf{u}$ is in H .

$A: m \times n$

Examples:

- Column space of a matrix A ($\text{Col } A$) contains all linear combinations of columns of A (subspace of $\mathbb{R}^m$)
- Null space of a matrix A ($\text{Nul } A$) contains all solutions of the homogeneous equation $A\vec{x} = \vec{0}$ (subspace of $\mathbb{R}^n$)

A **basis** for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

basis of $\text{Col } A$: pivot columns

basis of $\text{Nul } A$: from the parametric vector form.

dimension: number of vectors in a basis

Dimension and Rank

The **dimension** of a nonzero subspace H , denoted by $\dim H$, is the number of vectors in any basis for H . The dimension of the zero subspace $\{0\}$ is defined to be zero.²

The **rank** of a matrix A , denoted by $\text{rank } A$, is the dimension of the column space of A .

The Rank Theorem

If a matrix A has n columns, then $\text{rank } A + \dim \text{Nul } A = n$.

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

- m. The columns of A form a basis of $\mathbb{R}^n$. $\Rightarrow$ pivot in every column
- n. $\text{Col } A = \mathbb{R}^n \Rightarrow m$.
- o. $\text{rank } A = n \Rightarrow m$.
- p. $\dim \text{Nul } A = 0 \Rightarrow$ no free variable $\Rightarrow$ pivot in every column
- q. $\text{Nul } A = \{0\} \Rightarrow p$.

Alternating Sign Pattern:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \\ + & - & + & \\ \vdots & & & \ddots \end{bmatrix}$$

Introduction to Determinants

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\begin{aligned} \det A &= \underline{a_{11} \det A_{11}} - \underline{a_{12} \det A_{12}} + \cdots + \underline{(-1)^{1+n} a_{1n} \det A_{1n}} \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j} \end{aligned}$$

← cofactor expansion across the first row.

$$(i,j)\text{-cofactor} = C_{ij} = (-1)^{i+j} a_{ij} \det(A_{ij})$$

Theorems:

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

A square matrix A is invertible if and only if $\det A \neq 0$. (A is nonsingular)

Example of finding A_{ij}

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

then A_{32} is obtained by crossing out row 3 and column 2,

$$\begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix}$$

so that

$$A_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Properties of Determinants

$$(b) B = \begin{pmatrix} 0 & 1 & t \\ -3 & 10 & 0 \\ 0 & 5 & k \end{pmatrix} \xrightarrow{?} \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

Row Operations

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

If A is an $n \times n$ matrix, then $\det A^T = \det A$. $\Rightarrow$ row operations

Multiplicative Property

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

Now we have two general techniques for computing determinants of $n \times n$ matrix A ($n \geq 2$)

Example:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

[Method 1]: (Cofactor Expansion)

$$\det(A) = (-1)^{2+1} (-3) \det \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix} = 3 \cdot (1 - 10) = -27$$

[Method 2]: (Row Operations)

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -3 & 10 & 0 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{change sign}]{R_1 \leftrightarrow R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 5 & 1 \end{bmatrix} \xrightarrow[\text{no change}]{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix}$$

$$\det(A) = -\det \begin{bmatrix} -3 & 10 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -9 \end{bmatrix} = -1((-3) \times 1 \times (-9)) = -27$$

Cramer's Rule, Volume, and Linear Transformation

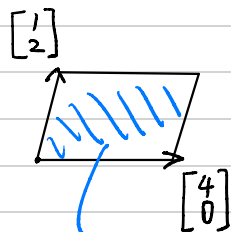
Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

Determinant as Area or Volume

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.



$$\text{Area} = \left| \det \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix} \right|$$

Linear Transformations

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

Application to Markov Chains

A vector with nonnegative entries that add up to 1 is called a **probability vector**.

A **stochastic matrix** is a square matrix whose columns are probability vectors.

A **Markov chain** is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots$, together with a stochastic matrix P , such that

$$\mathbf{x}_1 = P\mathbf{x}_0, \quad \mathbf{x}_2 = P\mathbf{x}_1, \quad \mathbf{x}_3 = P\mathbf{x}_2, \quad \dots$$

Thus the Markov chain is described by the first-order difference equation

$$\mathbf{x}_{k+1} = P\mathbf{x}_k \quad \text{for } k = 0, 1, 2, \dots$$

$\mathbf{x}_i$ is called a state vector $i=1, 2, \dots$

Regular stochastic matrix is when P^2 has every entry being positive.

Eigenvectors & Eigenvalues

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to λ* .¹

↑ Eigenvector
↗ Eigenvalue

$$A\vec{x} = \lambda\vec{x} \Rightarrow A\vec{x} - \lambda\vec{x} = 0 \Rightarrow (A - \lambda I)\vec{x} = 0$$

If there exists a nonzero eigenvector $\vec{x}$ corresponding to the eigenvalue λ then $\det(A - \lambda I) = 0$

The Characteristic Equation

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

The general idea of find eigenvalues and eigenvectors:

- ① define the characteristic eqn: $\det(A - \lambda I) = 0$
- ② Solve for the eigenvalues (solve the characteristic eqn).
 $\lambda_1, \dots, \lambda_n$
- ③ for each $\lambda_1, \dots, \lambda_n$ find the basis of the null space of $A - \lambda_i I$
 $[A - \lambda_1 I | 0], [A - \lambda_2 I | 0], \dots, [A - \lambda_n I]$